

Quantum-Aware Classically-Trained Embeddings via Projective Metric Learning

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Abstract

Quantum Machine Learning (QML) promises richer representations and improved learning by leveraging the unique properties of quantum computation. A necessary first step in using QML is to encode classical data into quantum states. Static encoding mechanisms offer limited expressivity, while quantum training suffers from barren plateaus, making optimization unstable and inefficient. We propose Quantum Projective Metric Learning (QPMeL) - a quantum-aware, classically-trained method to learn dense and high-quality quantum encodings. QPMeL achieves this by mapping classical data to the surface of independent unit spheres in $\mathbb{R}^3$, which naturally aligns with the state of multiple unentangled qubits. QPMeL also introduces a novel Projective Metric Function (PMeF) to approximate Hilbert space similarity in $\mathbb{R}^3$ and a gradient stabilization trick further enhances training efficiency. QPMeL achieves state-of-the-art performance on MNIST, Fashion-MNIST, and Omniglot, scaling up to 10-class classification and 15-way few-shot learning with high accuracy using significantly fewer qubits. It is also the first QML approach to support multi-modal (image-text) learning, achieving over 90% accuracy in the 15-way-1-shot setting with just 20 qubits.

Introduction

Quantum computing introduces a novel computational paradigm by exploiting quantum phenomena such as superposition and entanglement, enabling exponentially richer information processing than classical methods (Schuld 2021). In machine learning, these capabilities promise richer feature representations, improved function approximation, and faster learning. Consequently, Quantum Machine Learning (QML) is gaining attention for its potential to address complex tasks beyond the reach of classical models. Despite the promise of QML, practical solutions are severely constrained by the current limitations of quantum hardware. We are still in the Noisy Intermediate-Scale Quantum (NISQ) era, where available devices have limited qubit counts, short coherence times, and no robust error correction. Therefore, it is imperative to consider NISQ devices when designing QML workflows.

To apply quantum models to classical data, it needs to be encoded onto quantum states. The effectiveness of this encoding often determines the downstream performance of the

quantum model (Havlíček et al. 2019). The quality of encoding is typically measured in terms of the density of information, and the separability of similar data (ex: samples from the same class) in the encoding space. While better encodings yield better models, they only represent the starting point of the QML model and do not replace trainable circuits required for downstream tasks.

Early quantum data encoding techniques, such as amplitude and angle encoding were static (Mari et al. 2020). With classical pre-processing to compress data, these methods were too expensive in the number of qubits or number of gates to be practical for NISQ devices. When paired with pre-trained classical compression models such as PCA, they often yield sparse, poorly separated encodings as the classical model is trained independently of the quantum circuit. Trainable quantum encodings (Lloyd et al. 2020; Hur, Araujo, and Park 2023; Hou et al. 2023; Wang et al. 2024) were introduced to address these issues, allowing the quantum circuit to learn dense, task-optimized encodings directly in the quantum state space. However, the presence of barren plateaus (McClean et al. 2018; Cerezo et al. 2021) in the optimization landscape makes training these circuits difficult, slow, and resource-intensive. This limits their applicability on current quantum hardware.

In this work, we introduce **Quantum Projective Metric Learning (QPMeL)**—a novel framework that avoids both the challenges by enabling *quantum-aware, classically-*

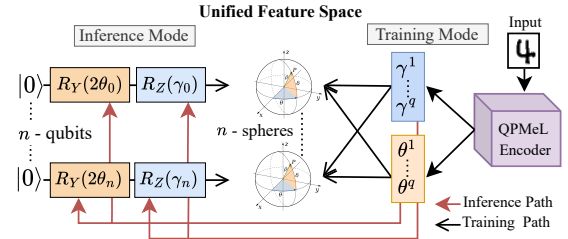


Figure 1: QPMeL utilizes the surface of independent unit spheres as a unified feature space between quantum states and classical vectors. This unified space spans all unentangled quantum states. Once trained, a NISQ friendly depth-efficient circuit using only 2 gates per qubit encodes data.

trained data encodings. The core idea of QPMeL is to define a unified feature space composed of the surfaces of independent unit spheres in $\mathbb{R}^3$. Classical data points are mapped to this space by a learnable encoder, producing polar and azimuthal angles (θ, γ) that naturally align with the state space of multiple unentangled qubits (think multiple Bloch spheres). These coordinates are directly translated into quantum states using only R_Y and R_Z gates. Due to only training classically, QPMeL avoids barren plateaus.

QPMeL introduces a novel *Projective Metric Function* (PMeF) which can compute the similarity in quantum state space using the $\mathbb{R}^3$ coordinates of the encoded points. Additionally, we introduce a gradient trick to improve training stability and convergence. The key contributions of QPMeL can be summarized as:

1. A unified feature space consisting of independent spherical surfaces common to the classical and quantum domains created via a classical encoder which outputs angular encodings (θ, γ) .
2. A novel Projective Metric Function (PMeF), which computes the similarity between points in quantum state space using only the $\mathbb{R}^3$ coordinates derived from the angular encodings.
3. A gradient trick for PMeF leading to more stable gradients during training, allowing the models to converge more consistently.

We benchmark QPMeL against a broad set of quantum metric learning approaches. In standard classification, QPMeL outperforms all prior methods on MNIST and Fashion-MNIST in binary, 3-class, and 10-class settings. It achieves nearly 99% accuracy on binary tasks, including harder class pairs like (3,5) and (3,6), where other methods degrade significantly. It is the only method that scales successfully to 10-class classification, reaching 96% on MNIST and 85% on Fashion-MNIST. Previous methods either support only binary classification (Hur, Araujo, and Park 2023; Hou et al. 2023), or report no multi-class results, or perform poorly on more complex tasks (Wang et al. 2024).

In few-shot learning (FSL), QPMeL again outperforms

or matches prior work (Huang, Shi, and Li 2025; Liu et al. 2022), while using only 1/6th the number of qubits. It scales to 10-way FSL with 90% accuracy and maintains over 85% accuracy in the 15-way setting, which most baselines cannot handle.

To the best of our knowledge, QPMeL is the first QML method to scale up to support multi-modal (image-text) few-shot learning. Using a CLIP-like architecture (BERT for text and Xception for image), QPMeL achieves $\geq 90\%$ accuracy in 15-way 1-shot classification. Its performance remains consistent across support/query modality combinations, suggesting it learns a shared latent space for both modalities. All of this is achieved using just 20 qubits, (compared to 64 for image-only models like Liu et al. (Liu et al. 2022)) highlighting QPMeL’s scalability and efficiency.

Background and Limitations of Related Works

A key challenge in Quantum Machine Learning (QML) is encoding classical data onto quantum states in a way that is both expressive and hardware-efficient. As demonstrated by Havlíček et al. (2019), the choice of encoding has a significant effect on the performance and learnability of quantum models. Given the limited number of qubits available on current NISQ devices, the encoding step must be both qubit-efficient and capable of producing task-relevant embeddings. Figure 2 illustrates four major families of encoding strategies based on how classical models interact with quantum circuits.

(i) Classical Compression + Static Quantum Encoding: Early works (Mari et al. 2020; Havlíček et al. 2019) proposed using classical dimensionality reduction techniques such as PCA or autoencoders to compress the data, which is then encoded into quantum states using static quantum circuits (e.g., angle or amplitude encoding). While these approaches are simple and hardware-friendly, the use of fixed single-axis encodings (like R_Y gates) limits expressivity. They typically access only a small subspace of the full Hilbert space, and do not even attempt to achieve class separability in the encoded quantum states.

(ii) Classical Compression + Trainable Quantum Encod-

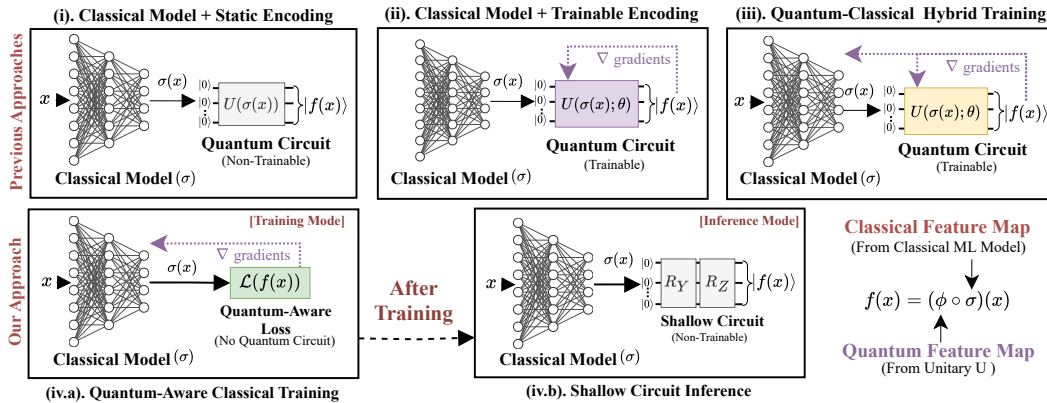


Figure 2: Main families of quantum data encodings classified based on the relationship between the classical model and quantum circuit.

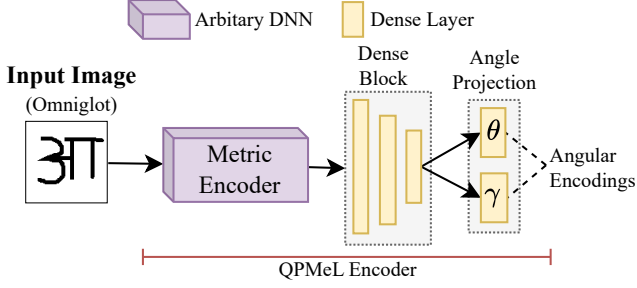


Figure 3: QPMeL Encoder: Consists of a standard metric encoder with a dense block appended to it. The final layer of the dense block is used as the input to 2 independent dense layers, which produce the angular coordinates $\vec{\theta}, \vec{\gamma}$. This is termed ‘Angle Projection’.

ing: To improve Hilbert space utilization, Lloyd et al. (2020) introduced Quantum Metric Learning (QMeL), where the compressed classical data is passed into a trainable quantum circuit. Variants of this approach include quantum random access codes (Thumwanit et al. 2021), which reduce qubit count but maintain trainability. While this class of models can, in principle, learn dense and task-specific embeddings, they require deep circuits and suffer from instability during training due to the non-Euclidean and periodic structure of quantum states.

(iii) Hybrid Classical + Quantum Training: Recent works (Hou et al. 2023; Liu et al. 2022; Huang, Shi, and Li 2025; Wang et al. 2024; Hur, Araujo, and Park 2023) propose training both the classical encoder and the quantum circuit jointly, allowing the classical model to adapt to the quantum feature space. This hybrid approach improves representation quality but comes at a cost: joint optimization is highly susceptible to the *barren plateau* problem (McClean et al. 2018; Cerezo et al. 2021), where gradients vanish as the number of qubits increases, leading to poor convergence and extremely slow training.

(iv) Quantum-Aware Classical Training (QPMeL): To overcome the limitations of the above methods, we propose QPMeL. In QPMeL a classical model learns a mapping to quantum space using a *quantum-aware loss function*, without requiring a trainable quantum circuit. The encoder is trained to produce angular parameters (θ, γ) that are trained to be separable when transformed into quantum states. This enables the model to learn dense, expressive embeddings that respect Hilbert space geometry while avoiding the training instability associated with quantum circuits. Once trained, only 2 gates per qubit are required to encode data, making it extremely depth-efficient for NISQ devices.

Quantum Projective Metric Learning

The Quantum Projective Metric Learning (QPMeL) is a framework designed to learn dense and well-separated quantum embeddings using a ‘Quantum-Aware’ loss function and fully classical training. QPMeL constructs a unified feature space comprising the surfaces of independent unit spheres, reflecting the Bloch sphere geometry of individual qubits.

QPMeL consists of 2 main components - (1). The QPMeL encoder, and (2). Projective Metric Function (PMeF).

1. QPMeL Encoder: The QPMeL encoder is a modified metric encoder which outputs 2 independent vectors $\vec{\theta}, \vec{\gamma}$, such that all values in $\vec{\theta}$ range from $[0, \pi]$ and all values in $\vec{\gamma}$ range from $[-\pi, \pi]$. As shown in Figure 3, this is done via adding 2 parallel dense layers to the end of the model, whose outputs are scaled to the desired ranges. This is termed ‘Angle Projection’. These vectors represent the polar and azimuthal angles in the polar coordinate systems and are therefore termed ‘Angular Coordinates’. A more detailed look is provided in the appendix.

2. Projective Metric Function: PMeF computes the similarity between 2 points in quantum state space using only their real-valued angular coordinates.

During inference, the angular coordinates from the QPMeL encoder are used to parameterize R_Y and R_Z gates respectively, allowing for low-depth, NISQ friendly, encodings. Critically, during training, QPMeL operates entirely in classical real-valued space, relying on the novel PMeF to compute similarity in quantum state space for metric loss functions (ex. Prototypical loss). This allows QPMeL to avoid the optimization issues associated with quantum training (e.g., barren plateaus) while maintaining awareness of quantum state space geometry.

Inference Pipeline

As shown in Figure 4, for a set of target qubits $q \in [1, Q]$, the fully classical QPMeL encoder produces 2 vectors of length Q containing the corresponding angular coordinates $(\vec{\theta}, \vec{\gamma})$. These coordinates represent the polar and azimuthal angles for each qubit. During inference, the desired state for each qubit is produced by parameterizing R_Y and R_Z gates with $2\theta_q \in \vec{\theta}$ and $\gamma_q \in \vec{\gamma}$, respectively. This produces a fully factorized Q -qubit quantum state:

$$|\phi\rangle = \bigotimes_{q=1}^n \cos(\theta_q) |0\rangle + e^{i\gamma_q} \sin(\theta_q) |1\rangle \quad (1)$$

The use of $R_Y(2\theta_q)$ instead of the conventional $R_Y(\theta_q)$ is intentional. The $R_Y(\theta_q)$ operation results in the state $\cos(\theta/2) |0\rangle + \sin(\theta/2) |1\rangle$, but the desired representation

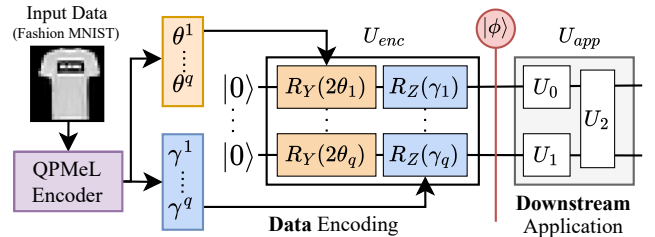


Figure 4: QPMeL Inference Pipeline: The angular coordinates from the QPMeL model are used to parameterize the quantum gates. The resulting quantum state is then measured to produce the final embedding.

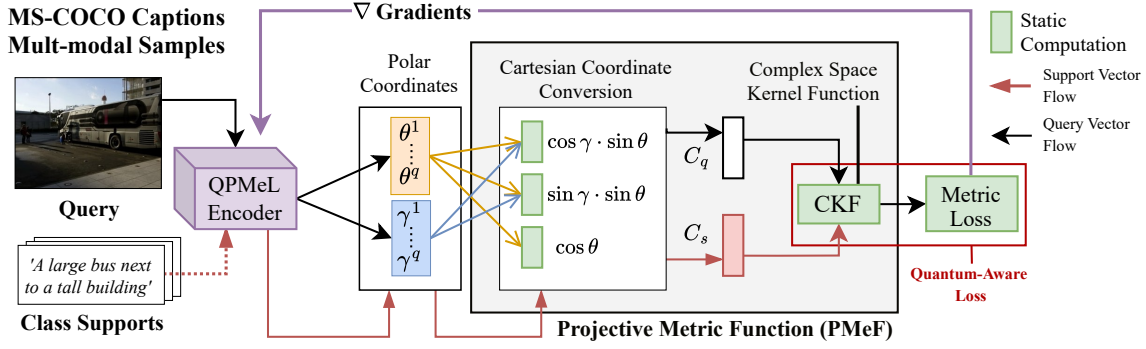


Figure 5: QPMeL training with prototypical loss: The QPMeL encoder produces a set of angular coordinates for the multi-modal support and query vectors. These angular coordinates are converted to Cartesian form, after which the Complex Kernel Function (CKF) is used to compute similarity to be used with a Metric Loss function. The combination of CKF and metric loss creates a ‘Quantum-Aware’ loss function. Cartesian Conversion and CKF jointly create the Projective Metric Function (PMeF).

in polar form is $\cos(\theta) |0\rangle + \sin(\theta) |1\rangle$. Using a double rotation angle (2θ) aligns the resulting state with this intended geometric interpretation. The global phase absorbs the denominator from the R_Z gate.

The resultant quantum state can then be used as the input for ‘Downstream Applications’ such as generative or regressive models. Due to only requiring a single R_Y and R_Z gate per qubit, QPMeL can be used with any circuit in place of static angle encoding, making it extremely flexible. Critically, the downstream application circuit may need to be trained separately based on the target application; however, no retraining of QPMeL is required. Downstream applications can employ any circuit architecture, including those with entanglement, as separability is only relevant for the encoding state, not the subsequent states.

Training Pipeline

QPMeL aims to train a classical model that generates dense angular encodings which map to well-separated, n -qubit states spanning the full quantum state space of unentangled states. Notably, QPMeL avoids using quantum circuits during training to avoid challenges inherent to training them.

QPMeL can be trained with any metric loss function such as *Prototypical loss* (Snell, Swersky, and Zemel 2017), which rely on computing the similarity between encodings to learn a well-separated encoding space. As shown in Figure 5, a small number of labeled examples (supports) are used to create a ‘proto-vectors’ (C_S) for each class. Query samples are then classified based on their similarity to these proto-vectors. The encodings are trained to maximize the similarity between the query and the proto-vectors of its corresponding class, while minimizing the similarity to proto-vectors of all other classes. In the few-shot training regime, the set of classes changes every batch. The resulting encoder produces discriminative and generalized encodings.

However, classical similarity functions such as cosine similarity or inner products do not properly reflect the nuances of quantum state space. Therefore, QPMeL introduces PMeF to alleviate this issue. The combination of PMeF with any metric loss creates a ‘Quantum-Aware’ loss function.

Projective Metric Function (PMeF)

Classical similarity metrics such as cosine similarity or inner products, when applied directly to angular encodings, do not accurately reflect the geometry of quantum state space. Quantum state space differs from the $\mathbb{R}^n$ -space that the outputs of classical models normally occupy in 2 key ways -

1. **Coordinate Periodicity:** Flat $\mathbb{R}^n$ space does not have coordinate periodicity, i.e. distinct coordinates always correspond to distinct points; however, in spherical quantum space, distinct coordinates can correspond to the same point due to rotational periodicity.
2. **Complex-Valued Coordinates:** Quantum states occupy $\mathbb{C}^n$ and therefore contain complex-valued coordinates as opposed to the real-valued classical vectors $\in \mathbb{R}^n$.

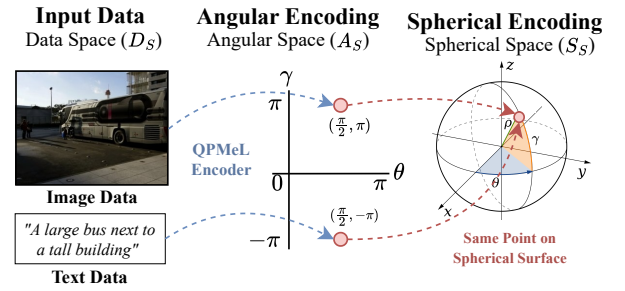


Figure 6: Spherical feature space of quantum states (S_S) vs flat $\mathbb{R}^2$ space of classical vectors (A_S). Points that might be distinct and far apart in A_S are the same point in S_S .

The Projective Metric Function (PMeF) addresses both issues using 2 main steps - (1). Polar-to-cartesian conversion, and (2). a Complex Kernel Function (CKF).

Step 1. Polar-to-Cartesian Conversion for Coordinate Periodicity: Classical $\mathbb{R}^n$ -space is flat (i.e. no periodicity in coordinates), whereas quantum states occupy a spherical surface and therefore have periodicity in their coordinates.

For example, consider angular encodings -

$$\vec{a} = (\theta_1 = \frac{\pi}{2}, \gamma_1 = \pi) \text{ and } \vec{b} = (\theta_2 = \frac{\pi}{2}, \gamma_2 = -\pi)$$

They both represent the same point on the Bloch sphere as they correspond to the same quantum state when inputted to equation 1 - $|\psi\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

However, they represent distinct and distant points in $\mathbb{R}^2$ as shown in Figure 6. Therefore, using cosine similarity directly on the angular encodings incorrectly indicates that the encodings are dissimilar, despite them corresponding to the exact same quantum state as shown below -

$$\begin{aligned} \cos(\text{angle}) &= \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|} = \frac{\theta_1 \theta_2 + \gamma_1 \gamma_2}{\|\vec{a}\| \|\vec{b}\|} \\ &= \frac{\frac{\pi^2}{4} - \pi^2}{\sqrt{\frac{5\pi^2}{4}} \cdot \sqrt{\frac{5\pi^2}{4}}} = -\frac{3}{5} \end{aligned}$$

PMeF addresses this by converting each pair of angular coordinates (θ, γ) into real-valued Cartesian coordinates on a $\mathbb{R}^3$ spherical surface as follows:

$$(\theta, \gamma) \rightarrow (x, y, z) = (\sin \theta \cos \gamma, \sin \theta \sin \gamma, \cos \theta) \quad (2)$$

Step 2. Complex Kernel Function: Quantum state space is comprised of exponentially large ‘complex-valued’ vectors. However, in order to efficiently learn mappings classically, we must be able to compute the similarity between quantum states using the **real-valued** angular encoding vectors which scale **linearly** in the number of target qubits. This is achieved via a novel ‘Complex Kernel Function’(CKF).

Consider a set of Q -length angular encodings $\{(\theta_q, \gamma_q)\}$ for qubits $q \in [1, Q]$. As described in equation 1, when encoded onto a quantum circuit they create the state :

$$\begin{aligned} |\psi\rangle &= \bigotimes_{q=1}^Q \cos(\theta_q) |0\rangle + e^{i\gamma_q} \sin(\theta_q) |1\rangle \\ &= \bigotimes_{q=1}^Q \cos(\theta_q) |0\rangle + (\sin \theta_q \cos \gamma_q + i \sin \theta_q \sin \gamma_q) |1\rangle \end{aligned} \quad (3)$$

We can write this state in terms of the real-valued cartesian coordinates introduced in equation 2 as

$$= \bigotimes_{q=1}^Q (z_q |0\rangle + (x_q + iy_q) |1\rangle) \quad (4)$$

Consider 2 quantum states $(|\psi\rangle, |\phi\rangle)$ produced from the angular encodings $\vec{A}_\psi, \vec{A}_\phi$. The similarity between the states is computed via quantum state fidelity $\mathcal{F}(|\psi\rangle, |\phi\rangle) = |\langle\psi|\phi\rangle|^2$. Due to QPMeL using unentangled factorized states, we can compute the fidelity between the multi-qubit states as the product of single-qubit fidelities (proof in appendix):

$$|\langle\psi|\phi\rangle|^2 = \prod_{q=1}^Q |\langle\psi_q|\phi_q\rangle|^2 \quad (5)$$

We can rewrite the states $|\psi_q\rangle$ and $|\phi_q\rangle$ in terms of the cartesian coordinates (x_q, y_q, z_q) and (x'_q, y'_q, z'_q) respectively using equation 4. Therefore, $|\langle\psi_q|\phi_q\rangle|^2$ can be computed using the permuted dot products of the cartesian coordinates:

$$|\langle\psi_q|\phi_q\rangle|^2 = (x_q x'_q + y_q y'_q + z_q z'_q)^2 + (x_q y'_q - y_q x'_q)^2$$

Collect the permuted dot products into 2 terms :

$$\lambda_q^r = x_q x'_q + y_q y'_q + z_q z'_q \text{ and } \lambda_q^c = x_q y'_q - y_q x'_q$$

These represent the real and imaginary components of the complex inner product and can be efficiently computed in linear time on classical computers via vectorized dot products (proof in the appendix).

Therefore, we can define a kernel function to compute the fidelity between 2 multi-qubit states as a function of λ_q^r and λ_q^c as follows:

Definition 1 (Complex Kernel Function (CKF)). *Given the single-qubit states $|\psi_q\rangle$ and $|\phi_q\rangle$, the CKF can be defined to compute $|\langle\psi_q|\phi_q\rangle|^2$ using the dot products (λ_q^r and λ_q^c) of their corresponding real-valued cartesian coordinates $\vec{C}_{\psi_q} = (x_q, y_q, z_q)$ and $\vec{C}_{\phi_q} = (x'_q, y'_q, z'_q)$ as follows:*

$$CKF(\vec{C}_{\psi_q}, \vec{C}_{\phi_q}) = (\lambda_q^r)^2 + (\lambda_q^c)^2 \quad (6)$$

We can now define PMeF by combining equation 2, equation 5, and equation 6.

Definition 2 (Projective Metric Function (PMeF)). *PMeF computes the fidelity between states $|\psi\rangle, |\phi\rangle$ via their angle encodings $\vec{A}_\psi, \vec{A}_\phi$ as follows:*

$$PMeF(\vec{A}_\psi, \vec{A}_\phi) = \prod_{q=1}^Q [(\lambda_q^r)^2 + (\lambda_q^c)^2] \quad (7)$$

Therefore, PMeF computes multi-qubit fidelity in **linear time** using only the **real-valued** angular encodings.

Gradient Trick The multiplicative structure of PMeF outlined in equation 7, can lead to vanishing gradients during training—especially as Q increases (proof in the appendix). To mitigate this, we modify the metric used during training by replacing the product with a sum over single-qubit fidelities:

$$PMeF_{\text{train}}(\psi, \phi) = \sum_{q=1}^Q |\langle\psi_q|\phi_q\rangle|^2 \quad (8)$$

This additive form retains the optimization direction (minimizing the similarity between dissimilar samples and maximizing it for similar ones), but avoids the exponential shrinking of gradients. This trick leads to more stable and faster convergence, especially for higher-qubit settings.

Experimental Setup

Datasets Selected: We evaluated QPMeL on 4 datasets primarily on two tasks - (1). *Standard classification* where inputs are sampled for a fixed set of classes, for example MNIST (0,1) only samples images from the classes

Table 1: Standard classification accuracy against SOTA

Dataset	Approach	# Qubits	Binary	3-Class	10 - Class
MNIST	Hur, Araujo, and Park (2023)	12	99.2 ± 0.4 (0,1)	✗	✗
	Hou et al. (2023)	6	99.0 (0,1)	✗	✗
	Thumwanit et al. (2021)	9	91.7 (3,6)	✗	✗
	Huang, Shi, and Li (2025)	9	88.8 ± 8.6 (0,1)	$\emptyset$	$\emptyset$
		9	74.7 ± 4.6 (3,5)	$\emptyset$	$\emptyset$
	Wang et al. (2024)	6	96.5 (0,1)	50.5 (0,1,2)	$\emptyset$
	QPMeL (Ours)	5	99.8 ± 0.1 (0,1) 99.6 ± 0.1 (3,6) 98.2 ± 0.4 (3,5)	99.1 ± 0.5 (0,1,2) 98.9 ± 0.4 (3,5,6) -	96.3 ± 0.3 - -
Fashion MNIST	Hur, Araujo, and Park (2023)	12	95.5 ± 0.1 (0,1)	✗	✗
	Huang, Shi, and Li (2025)	9	80.9 ± 9.8 (0,1)	$\emptyset$	$\emptyset$
	Wang et al. (2024)	6	89.8 (0,1)	48.3 (0,1,2)	$\emptyset$
	QPMeL (Ours)	6	98.0 ± 0.6 (0,1)	95.6 ± 0.5 (0,1,2)	85.0 ± 0.3

✗ = Approach cannot work for the given task, $\emptyset$ = No results provided in the original paper

0 and 1 to test accuracy. **MNIST** (Deng 2012), **Fashion-MNIST** (Xiao, Rasul, and Vollgraf 2017) are used to evaluate standard classification performance. (2). *Few-Shot Classification* where sets of inputs are sampled from all available classes, for example, Omniglot 2-Way samples different random classes each time from all available classes. **Omniglot** (Lake, Salakhutdinov, and Tenenbaum 2015), and **MS-COCO Captions** (Chen et al. 2015) are all tested on few-shot classification.

Testing Methodology: All of the results for QPMeL were computed using the ‘test’ set provided by the dataset unless otherwise specified. Each experiment was repeated 150 times to compute the 95% confidence interval reported as *mean $\pm$ interval* in all tables. Additional details on the experimental setup are provided in the appendix.

Baselines: We compare against all the quantum metric learning papers (Hur, Araujo, and Park 2023; Hou et al. 2023; Thumwanit et al. 2021; Huang, Shi, and Li 2025; Wang et al. 2024; Liu et al. 2022), except Lloyd et al. (2020), whose code was retracted and numerical results were not provided in the original paper. Due to hyper-parameter sensitive nature of training QML models and lack of publicly available code or pre-trained models, reproducing the results of the baseline approaches presents a challenge. Therefore, we take the accuracy numbers directly from their respective papers.

Results

QPMeL significantly outperforms previous methods in Standard Classification

Table 1 highlights the performance of QPMeL as compared to existing QMeL approaches. The first and second columns list the dataset and approach being tested respectively. Columns 3 lists the number of qubits required by the respective approach. Finally, columns 4,5 and 6 present the binary, 3-class and 10-class classification accuracies respectively. The binary and 3-class accuracies are reported

with the sampled classes specified in brackets. For example, (0, 1, 2) indicates that the accuracy is computed by sampling from classes 0,1 and 2.

We can see from column 4 that QPMeL outperforms all other approaches on the MNIST and Fashion-MNIST datasets in Binary Classification, achieving a 99% accuracy. The difference in accuracies between the (0,1) and (3,5) sets for Huang, Shi, and Li (2025), highlights that the choice of classes can significantly impact the classification accuracy of a given model. While most of the previous approaches (Hur, Araujo, and Park 2023; Hou et al. 2023; Wang et al. 2024) only reported results for the simpler (0,1) task, QPMeL achieves $> 98\%$ accuracy even on the more difficult (3,5) and (3,6) sets. Table 1 details QPMeL’s performance in a few shot setting with randomly sampled classes, which eliminates the bias introduced by the choice of classes. Additionally, Hur, Araujo, and Park (2023); Thumwanit et al. (2021) and Hou et al. (2023) cannot scale (denoted by ✗) beyond binary classification as they output binary labels with no details on multi-class extensions. Meanwhile, Huang, Shi, and Li (2025) use distance based classification and could be adapted to multi-class settings but do not provide any results in their work (denoted by $\emptyset$). QPMeL significantly outperforms Wang et al. (2024), achieving $\sim 2\times$ increase in 3-Class classification on the (0, 1, 2) and (3, 5, 6) sets as shown in column 5. Column 6 highlights that QPMeL is the only approach to scale upto 10-class classification on both datasets while achieving $\sim 96\%$ accuracy on MNIST and $\sim 85\%$ on Fashion-MNIST.

The significant improvement in accuracies using QPMeL can be attributed to the improved training of the classical model due to better gradients. QPMeL is able to train classical models to learn mappings to quantum features space directly via the PMeF, avoiding those issues, leading to better training and learning a more efficient mapping from data to Hilbert Space.

Table 2: N-Way-5-shot FSL performance

Dataset	Approach	#Qubits	2-Way	5-Way	10-Way
Omniglot	Huang, Shi, and Li (2025)	12	87.39*	$\emptyset$	$\emptyset$
	Liu et al. (2022)	64	$\emptyset$	98.0	$\emptyset$
	QPMeL (Ours)	11	98.13 $\pm$ 0.49	96.0 $\pm$ 0.5	90.02 $\pm$ 0.42

* Only sampled from 2 fixed classes, $\emptyset$ = No results provided in the original paper

QPMeL significantly outperforms previous methods in FSL classification

Table 2 presents the FSL performance of QPMeL compared to Huang, Shi, and Li (2025) and Liu et al. (2022). The table uses a similar structure to Table 1, but columns 4,5,6 present the n -way classification performance where support and queries are sampled from n randomly selected classes. Columns 2 and 4 highlight the baseline approach and the the number of qubits they utilize respectively. QPMeL **shows a 10% improvement in 2-Way accuracy** compared to Huang, Shi, and Li (2025), and in the 5-way setting is competitive with Liu et al. (2022), while **using only $\sim 1/6$ th the number of qubits**. Even when scaling to the 10-way setting, QPMeL achieves $\sim 90\%$ accuracy. Additional results in the appendix, show that QPMeL scales to the 15-way setting while still maintaining 85% accuracy.

QPMeL achieves strong Multi-Modal FSL classification performance

Table 3 presents the multi-modal 1-shot classification performance of QPMeL. The first column defines the number of classes or n -ways for few shot classification. The second and third columns represent the Image support - Text query and Text support - Image query configurations respectively. The support vectors are examples from each class for similarity comparison and the query vectors, are the new samples the model tries to classify. For each configuration, the sub-columns present the train set accuracy in both the classical setting (similarity computed via the PMeF) and quantum setting (similarity is computed via Inversion Test).

The core of QPMeL is the combination of the angular encodings created by the angle encoder and the similarity computed via the PMeF. This allows QPMeL to be integrated into any similarity/distance based learning framework, including multi-modal examples such as Radford et al.

(2021). QPMeL was used to train a CLIP-like model using a BERT(Vaswani et al. 2023) based language encoder and Xception Net(Chollet 2017) based image encoder. As shown in the last row of table 3 QPMeL achieves $> 90\%$ accuracy on the train-set even when scaling to 15-way-1-shot settings. Additionally, as we can see from the contrast between the Image-Text and Text-Image columns the choice of modality for support and query does not affect the performance of QPMeL. This is a strong indicator that QPMeL is able to learn a common representation space for both text and image data. This is a significant result as it indicates that QPMeL can be used to learn multi-modal embeddings in quantum space. Remarkably, QPMeL achieves this feat **using only 20 qubits**, while previous methods such as Liu et al. (2022) require 64 qubits for only the image modality. **We have not come across any previous methods showcasing multi-modal embeddings for quantum computers.** Additionally, notice that there exist no significant differences between the classical and quantum settings. This provides strong evidence that the model is learning a common representation space between the quantum and classical domains.

Conclusion and Limitations

In this paper we introduce QPMeL, a novel framework that learns encodings for quantum computer classically via a unified feature space of independent spherical surfaces and a novel Projective Metric Function (PMeF). Our results prove that QPMeL is more efficient and learns better representations than existing QML methods. While QPMeL may be limited by not being able to learn entangled states, our results show that this does not seem to affect the performance of QPMeL.

Table 3: FSL-1-shot Accuracy on MS-COCO Captions (20 qubits)

N-Way	Image - Text*		Text - Image*	
	Classical	Quantum	Classical	Quantum
5-Way	98.53 $\pm$ 0.92	98.80 $\pm$ 1.36	98.53 $\pm$ 0.84	98.00 $\pm$ 1.72
9-Way	98.07 $\pm$ 0.77	95.33 $\pm$ 2.12	97.04 $\pm$ 1.01	95.11 $\pm$ 2.40
13-Way	95.85 $\pm$ 0.92	95.23 $\pm$ 1.76	95.59 $\pm$ 0.95	93.69 $\pm$ 2.24
15-Way	94.71 $\pm$ 0.98	95.87 $\pm$ 0.88	93.60 $\pm$ 1.87	93.07 $\pm$ 1.71

*Support Modality - Query Modality.

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